Author Response: TriPoseFusion (Submission #802)

Method	MPJPE	PA-MPJPE	PCK@.05	PCK@.10
A. Ours, scale-anchored ref., fold-0 val, masked 52 joints (m)				
best view (right)	0.191	0.066	0.31	0.43
mean fusion	0.189	0.063	0.32	0.45
submitted full model	0.189	0.063	0.31	0.45
B. Ours, scale-anchored ref., 5-fold cross-subject (m)				
mean fusion	0.149	0.044	0.34	0.50
residual head, 8K	0.128	0.040	0.37	0.55
submitted arch., 870K	0.142	0.044	0.35	0.52
C. D&A, official split-0 ref., zero-shot, 10 joints (mm): rigid†/PA				
mean fusion (best view: 29.0/24.0)	31.6	23.1	–	–
submitted full model	32.5	23.7	–	–

Table 1. Corrected results (†rigid = rotation+translation). Separate sanity check: official-calibration DLT, 14 joints: 32.5 mm absolute MPJPE.

To all reviewers. We thank the reviewers. Auditing the full evaluation stack after R-LAYQ's Tab. 3 anomaly, we found four defects: (i) transposed rotation and norm-ratio scale in the Procrustes alignment; (ii) a zero-gradient gate-entropy term; (iii) a degenerate shoulder fallback collapsing whole frames; (iv) uncalibrated, constructed extrinsics in the triangulation reference (scale error, median 2.4×). We corrected (i), (iii), and (iv), and removed (ii). The reference is rebuilt by per-sequence bundle adjustment (reprojection $9.60 \rightarrow 3.31$ px). Because its free scale is anchored to SAM3D-predicted shoulder width, we treat PA-MPJPE and reprojection as primary and make no absolute-accuracy claim on our data. Tab. 1 gives corrected results (one protocol and valid-joint mask per block).

R-LAYQ W1 — metric anomaly. Cause: (i)+(iii); a synthetic regression test recovers the 0.04 m truth (old code: 0.7–2.4 m). Corrected, same masks: canonicalized mean fusion 0.971/0.564 m on the original reference (reversal gone); calibrated: Tab. 1A.

W2 — fallback rule. Missing rule: shoulder axis of the nearest earlier valid frame (a leading invalid prefix uses the first valid frame — the only look-ahead); none valid → canonical $+x$ axis; axis-parallel guard; pseudocode in §3.3.

W3 — independent reference. No independent MoCap subset was available; we drop the absolute-accuracy claim and bound the reference by validity/reprojection statistics (19jf W1) and sensitivity (conclusions unchanged across original/calibrated). Drive&Act (A2) is cross-rig transfer evidence only — its 3D reference is itself OpenPose-triangulated.

W4 — narrow the learned-component claims. We do: the submitted model matches mean fusion at the reported MPJPE/PA precision (Tab. 1A; A3). We narrow the contribution to calibration-free canonicalization + uniform fusion, with a modest clean-input gain (19jf W2); A1 is a diagnostic corruption test, not a new contribution. Efficiency (0.87M params, 6.2 ms/frame CPU) is fusion-stage only; end-to-end latency incl. SAM3D is unmeasured. Minor items (mapping, frame-rate, splits, imbalance) added; joint-limit evaluation: future work.

R-19jf W1 — reference reliability under occlusion. Joints with < 2 views passing the 40 px reprojection filter are masked out for *every* method. Validity: head/shoulder–neck 1.00, hands 0.34; hands dominate the error (0.316 m vs. torso 0.067) and are reported separately (plus a torso-only protocol). Reference construction uses manually checked 2D inputs, automatic DLT, filtering, and per-sequence bundle adjustment; uncertainty is reported.

W2 — canonicalization mechanism. Canonicalization defines the common frame enabling calibration-free fusion of per-view outputs that live in different camera frames; a raw-vs-canonicalized MPJPE is thus not meaningful (the paper's 2.078 m raw figure measured frame mismatch, not pose error). Its measurable effect is the fusion gain over the best view: small on clean input (Tab. 1A: MPJPE −1%, PA −4%, PCK@.10 +2 pts; Drive&Act PA 24.0→23.1), redundancy under view loss (dropped view: MPJPE unchanged), and calibration-free cross-rig transfer — a modest contribution, stated so in the revision. Module necessity: W4, A1.

R-xS2o A1 — gating collapse (W1–W2). We agree the stated entropy term, if active, favors uniform weights. Its implemented gradient was exactly zero (so it did not cause the collapse); retraining without it, with the fix, and with a leave-one-view-out teacher still gave 0.3333 gates — the median pseudo-teacher's trivial optimum. We remove the term and narrow the clean-input claim (W2 confirmed). In a diagnostic corruption test, a 1.9K gated head trained with corruption assigns 0.037 weight to a zeroed view; its MPJPE rises only 1–5%, versus 18% for mean fusion, while missing views receive zero weight.

A2 — Drive&Act (request 2). Protocol: official split-0 test subjects, no Drive&Act training; the 10 joints shared with BODY-25; PA-MPJPE main (no hips for canonical alignment) and rigid-MPJPE (Tab. 1C). Fusion beats every single view in PA; rigid-MPJPE also carries SAM3D's per-view scale bias (default shape ≈18% too large) that averaging does not remove. Published results use different joints/metrics; our zero-shot errors are only of the same order as the cited supervised 22.6–30.4 mm. The 0.412 m result was dominated by defect (iv)'s scale artifact (W3). Revision adds SAM3D details and PCK@.05/.10 on our data; reference limits: W3, 19jf W1.

A3 — deep-learning baselines (request 3). The requested UPose3D/LT/Faster VoxelPose baselines require calibrated 2D, image, or volume inputs, unlike our uncalibrated 3D-keypoint input. We therefore tested input-matched learned heads: LT-style confidence weighting (1.9K), uncertainty-aware set-attention (28K), residual (8K), and the submitted architecture (870K). Tab. 1B compares mean, the best head (8K), and our architecture under the same 5-fold protocol.